

Product Requirements Document: Replicate This (v2.0)

Project Name: Replicate This

Organization: FORRT (Framework for Open and Reproducible Research Training) / Replication Journal Federation

Status: V2.0 (Full-Stack Web Application)

Architecture: Serverless / Next.js / Supabase (Frankfurt Region)

1. Executive Summary

Replicate This is a community-driven, public registry for nominating published research papers that merit independent verification through replication or reproduction. This platform aims to change the scientific culture of criticism into one of collaboration by normalizing verification as a routine, shared scientific interest.

The 2.0 version transitions from a static registry to an interactive platform where researchers can claim studies, form collaborative working groups, and signal community interest through Reddit-style voting.

2. Objectives & Goals

- **Destigmatize Replication:** Frame nominations as an honor and recognition of a study's scientific importance.
- **Facilitate Collaboration:** Provide a "handshake" mechanism for researchers to "claim" studies and form working groups.
- **Maintain Zero-PII Compliance:** Adhere to strict German *Datenschutz* (GDPR) by ensuring no sensitive personal data (names/emails) is stored in the database.
- **Crowdsource Priorities:** Allow the academic community to highlight which papers are most urgently in need of replication through an upvote/downvote system.
- **Zero Infrastructure Costs:** Utilize a serverless architecture to ensure the platform requires near \$0 to host at scale.

3. User Personas

1. **The Nominator:** A researcher or student who identifies an influential study. They approve "Claimants" to form collaborative groups for their nominated study.
2. **The Replicator:** A researcher looking for a meaningful project. They browse the registry, claim studies, and join groups to coordinate verification efforts.
3. **The Maintainer (Admin):** A FORRT team member who reviews incoming nominations for adherence to the Code of Conduct and approves them for public display.

4. Functional Requirements

4.1 Authentication (Privacy-First)

- **Providers:** Exclusive support for Google, GitHub, and **ORCID** login.
- **No PII Storage:** Better Auth is configured to strip names and emails before saving to the database. Only a non-identifying UID and the provider's unique ID are persisted.
- **Session Management:** Secure, server-only cookies via Better Auth for high security and GDPR compliance.

4.2 The Nomination & Moderation Pipeline

- **Submission:** Users submit nominations via a structured form (DOI, Justification, Discipline).
- **Moderation Buffer:** Nominations default to a "Pending" state and are hidden from the public until approved by a Maintainer.
- **Status Management:** Administrators can label issues as "claimed" or "seeking collaborators" to update the public UI.

4.3 Collaboration & Grouping

- **Claiming:** Logged-in users can "claim" a study to signal intent to replicate.
- **Group Handshake:** The Nominator must accept a Replicator's claim to form a "Working Group."
- **Collaboration Hub:** A private, dynamic view accessible only to group members for internal communication and project coordination.
- **Inquiry System:** Third parties can message the Nominator/Replicator through a proxied form to protect user privacy.

4.4 Community Engagement

- **Reddit-Style Voting:** A standard upvote/downvote system on each nomination to signal scientific priority.
 - **Reputation (Optional):** Weighting votes from ORCID-verified accounts higher to ensure academic integrity.
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5. Technical Architecture

Component	Technology	Reasoning
Framework	Next.js 15+ (App Router)	High performance, SSR for privacy, and Server Actions for logic.
Auth	Better Auth	Type-safe, customizable, and handles ORCID via OIDC.
Database	Supabase (Postgres)	Hosted in Frankfurt (eu-central-1) for German compliance.
Deployment	Vercel	Scalable serverless hosting with zero initial cost.

6. Compliance & Privacy (*Datenschutz*)

1. **Data Residency:** All database clusters are pinned to the Frankfurt region.
2. **Minimization:** The application operates on "Identity-at-a-Distance." Personal profiles are fetched from providers at runtime and never stored locally.
3. **Right to Erasure:** A single-click "Delete Account" function that purges the hashed UID and all associated claims instantly.